
Optimizing Vision Transformers for White Shark Re-Identification

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1 Introduction

In this work, we examine the problem of white shark re-identification (Re-ID) from dorsal fin images Anderson et al. [2011]. Animal Re-ID, or matching anonymous data against a catalog of known individuals, is essential for understanding population dynamics and informing conservation efforts. Crucially, we formulate this as a retrieval, rather than a classification problem, as in many ecological monitoring scenarios, the set of identities is dynamic with new individuals discovered and others lost over time. Applying computer vision to automate this task has received immense attention Bergler et al. [2021], Blount et al. [2022], Hillman et al. [2002], Hughes and Burghardt [2016], Crall et al. [2013], Maroen et al. [2024], Van Tienhoven et al. [2007], Moskvyyak et al. [2021], Nepovinnikh et al. [2024], Parham et al. [2018], Thompson et al. [2021], Weideman et al. [2017], Zheng et al. [2022]. To succeed in the white shark domain, an image model must be highly sensitive to biomarkers on the dorsal fin, such as unique notches and pigmentation patterns, while remaining invariant to confounding variables such as pose, lighting, and image quality. Additionally, white shark sightings are rare and photographic conditions are often poor due to water quality and lighting. Thus, researchers can typically collect only a small number of quality photographs per individual, leading to a sparsely populated database of many individuals, each represented by few images.

We address these challenges and develop a framework for identifying individual white sharks from dorsal fin photographs leveraging a state-of-the-art vision transformer (ViT) architecture. Our model and retrieval system streamlines Re-ID by providing marine biologists a ranked list of suggested matching individuals for any query image, significantly reducing human involvement in labeling newly captured shark images. Our key contributions are: (1) training an image model optimized for our task by introducing class-aware triplet sampling and augmented images to improve performance with limited, sparsely distributed data; (2) evaluating two retrieval strategies, nearest-neighbor and prototype-based, for identifying individuals; and (3) demonstrating strong performance on an extensive dataset of dorsal fin images, achieving up to 60% hits@1 and 78% hits@10.

2 Method

ViT models pre-trained on large-scale datasets have been shown to provide state of the art results across a range of vision tasks, largely due to their flexible self-attention mechanisms. For our implementation, we selected the google/vit-large-patch16-384 model as a feature extraction backbone Dosovitskiy et al. [2021]. To adapt this foundation model to our specific task, we append a multilayer perceptron (MLP) projection head, trained to produce the final embeddings.

Model Training: To improve low performance observed with the baseline ViT backbone, we utilized low-rank adaptation (LoRA) Hu et al. [2021]. This method introduces a small set of trainable

parameters, enabling domain-specific adaptation without the computational cost of full fine-tuning. LoRA enabled model convergence within 72 hours on a single NVIDIA Quadro RTX 6000 GPU and resulted in a marked improvement in the model’s ability to generate discriminative embeddings, evidenced by significant gains in retrieval performance.

To train, we employ a triplet loss function Schroff et al. [2015] which aims to minimize the embedding space distance between an anchor/positive image pair (images of the same shark) while maximizing the distance between the anchor/negative pair (images of a different shark). Due to the sparse nature of our dataset where most individuals have few images, random batch sampling rarely yields triplets with the necessary positive pairs. We therefore implemented a class-aware triplet sampler to explicitly construct training batches where for each anchor image, at least two corresponding positive samples are included. This optimization supports the formation of valid triplets, increasing the model’s ability to learn effectively from each batch.

Evaluation following LoRA and class-aware triplet sampling showed our model learned to rely on spurious correlations, often grouping images based on irrelevant features such as background color or camera angle rather than the desired biomarkers. To address these issues, we introduced data augmentation. Generating variations of each training image by applying augmentations to perspective, rotation, scale, and color allowed us to artificially increase dataset size, address the scarcity of positive pairings, and provide explicit training to ignore differences in color, rotation, or obstruction of the fin.

Retrieval Techniques Sharks with the highest matching probability are retrieved by searching the embedding space for the nearest neighbors to a query image. A well-trained model should organize the embeddings into distinct clusters, each corresponding to a shark. Motivated by this, we compared two retrieval strategies: (1) a standard nearest neighbor approach which ranks potentially matching individuals based on the labels of the closest images in embedding space; (2) a prototype-based approach which computes a mean embedding for each individual using all of its available images. Matches are then ranked on proximity of these prototypes to the query embedding.

Model	Nearest Neighbor Retrieval			Prototype Retrieval		
	Hits@1	Hits@5	Hits@10	Hits@1	Hits@5	Hits@10
ViT	0.03	0.09	0.13	0.05	0.13	0.18
ViT+LoRA	0.09	0.23	0.29	0.11	0.23	0.30
ViT+LoRA+Samp.	0.31	0.55	0.64	0.33	0.56	0.64
ViT+LoRA+Samp.+Aug.	0.48	0.68	0.76	0.51	0.68	0.75

Table 1: Hits@K results for model variants comparing nearest neighbor and prototype retrieval.

3 Evaluation

We evaluated our models on a dataset of 3,083 dorsal fin images of 1,031 unique white sharks, curated by the Block Lab at Stanford University Andrzejczek et al. [2022]. We held out one image from each of the 497 individuals with two or more photos, resulting in train and test sets of 2,586 and 497 images, respectively. Experimental hits@K results, showing the proportion of queries for which the correct individual is present within the top-K retrieved matches, are presented in Table 1. Results demonstrate the progressive improvement in retrieval accuracy with each model optimization and the consistent superiority of prototype-based retrieval.

4 Conclusion

In this work, we develop and refine a pipeline for white shark Re-ID. We present optimizations applied to model architecture, training, and retrieval technique and show their benefits to retrieval accuracy. Our final pipeline has become a valuable tool for dataset de-duplication and matching newly captured shark images for our collaborators at the Block Lab. As future work, we also aim to refine the framework through further optimizations to both embedding generation and retrieval techniques. We also intend to address a limitation of our current framework by evaluating on additional datasets featuring different species to better understand its generalization potential.

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